

Safa Hichri

Cybersecurity & AI Engineering Student | Applied Cryptography · IAM · Secure Code

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📅 6-month internship from February 2027 · Open to relocation in Europe [🔗 Portfolio](#)

SUMMARY

Cybersecurity engineering student and AI master's candidate securing code, identities and AI agents. My master's thesis detects cryptographic API misuse in source code with an agentic system built on formal CrySL/CWE rules; I also built a security-hardened LLM agent for SOC automation (0.00% attack success rate against prompt injection and RAG poisoning) and a Zero Trust IAM architecture with SSO/MFA.

TECHNICAL SKILLS

AI & AI Security: LLMs, Agentic AI, Multi-agent Systems, RAG, LangChain, LangGraph, Ollama, OWASP LLM Top 10, Prompt Injection Defense, Machine Learning, Computer Vision | **Security Operations:** Wazuh (SIEM), SOAR, Threat Detection, Incident Response, Vulnerability Analysis, Nmap, Wireshark, Kali Linux | **Identity & Access:** IAM, Zero Trust, RBAC, Keycloak (SSO/MFA), OpenLDAP, Active Directory, HashiCorp Vault, AWS IAM | **Cryptography:** AES (ECB, CBC, GCM), RSA, ECC, Diffie-Hellman/ECDH, Hashing & Digital Signatures, CrySL, CWE | **Forensics:** Autopsy, Volatility 3, Scalpel, iLEAPP/ALEAPP | **Engineering & Systems:** Python, PowerShell, Git/GitHub, GitHub Actions (CI), Linux, Windows Server, TCP/IP, Cisco Networking, AWS, VMware

PROFESSIONAL EXPERIENCE

Master's Thesis (Ongoing)

06/2026 – Present | Sousse, Tunisia

[ToCodePro](#) [🔗](#)

Agentic Architecture for Cryptographic Misuse Detection in Source Code :

- **Designed** an agentic architecture combining **Agentic RAG** over a structured knowledge base of formal cryptographic rules (**CrySL/CWE**) with a **conditional multi-agent consensus mechanism** for cryptographic misuse detection.
- **Defined** a research positioning at the intersection of Agentic RAG, multi-agent consensus, and cryptographic API misuse detection, addressing an identified gap in existing approaches.
- **Structured** the system architecture to integrate specialized knowledge retrieval, source-code analysis, confidence-based decision making, and conditional multi-agent activation.

Bachelor's Final Year Project

02/2024 – 06/2024 | Sousse, Tunisia

[Future Vision Association](#) [🔗](#)

Real-Time Facial Recognition Access Control :

- **Deployed** a real-time **facial recognition access-control** system (92% accuracy) with liveness detection against identity spoofing and automated access logging.

EDUCATION

Engineering Cycle — Cybersecurity (2nd year, in progress)

2025 – Present | Sousse, Tunisia

[École Polytechnique de Sousse](#) [🔗](#)

Relevant coursework: **Cryptography**, Network Security, Artificial Intelligence .

Master's M2 — Artificial Intelligence

2024 – Present | Sousse, Tunisia

[ISSAT Sousse](#) [🔗](#)

PROJECTS

Zero Trust IAM Architecture & Operational Security — SecureCorp (Academic Group Project) [🔗](#)

- **Designed** a 3-VM Zero Trust architecture enforcing "Never Trust, Always Verify": identity (**OpenLDAP + Keycloak SSO/MFA**), **secrets management** and SIEM (**Vault + Wazuh**), and a secure access gateway (Nginx + Apache Guacamole).
- **Implemented RBAC across three roles** with brute-force protection, enforcing **least privilege** and zero plaintext passwords.
- **Architected** the authentication flow (**User → Nginx → Keycloak → RBAC → Vault → Resource**) with centralized **audit logging** in Wazuh.

AI-Powered SOC — Security-Hardened LLM Agent for Threat Detection & Response

- **Hardened** an LLM agent with three guard layers against prompt injection, RAG poisoning and tool misuse, reaching a **0.00% attack success rate** on held-out benchmarks (BIPIA).
- **Enforced** human sign-off on every action through RBAC and content-bound confirmation tokens. Ran **177 automated tests** in CI (GitHub Actions).

Digital Forensics Investigations — Disk, Memory, Network & Mobile Analysis

Correlated Wireshark, Autopsy and Volatility 3 evidence to reconstruct a full Windows user scenario. Identified a ransomware C2 server and recovered 3,221 deleted files.

ORGANISATIONS

IEEE ADACIS'25

Sousse , Tunisia

Co-organized IEEE ADACIS'25, a 3-day international conference featuring 6 keynote speakers from institutions including Imperial College London and the University of Quebec — **managing** venue logistics, registration for **150+ attendees** and event communications

Event Leader — Hack to the Future 2.0 CTF (Securinets EPS × IEEE CS)

Sousse , Tunisia

Organized a 70-participant CTF with multi-category challenges and technical workshops.

IA For All training & hackathon event

Sousse, Tunisia

Co-organized "IA For All," a 5-day AI training & hackathon event (ISSAT Sousse + Pristini School of AI) — coordinating 21 hours of expert-led training, a 24-hour hackathon with pitch sessions, and 5 partner organizations, for **55 participants**.